LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Hydraulic system, general design requirements, Typical layout of a basic hydraulic system and its brief description
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Hydraulic system, general design requirements, Typical layout of a basic hydraulic system and its brief description
5. Specific Objectives	23.1.2.1 Trainee will be able to describe typical hydraulic system and construction of its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of hydraulic component diagrams; Safety: No live systems; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a hydraulic system?
(b) Name one advantage of hydraulic power transmission over mechanical.
(c) What property of fluid is essential for hydraulic operation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General design requirements for hydraulic systems • Selection of working fluid and pressure ratings • Material compatibility and sealing requirements • Efficiency factors and heat dissipation needs • Safety standards and system redundancy	L & Ds	PPT, IB	15 min
(b)	• Pump types and their construction features • Actuator cylinders and motor construction • Control valves and directional logic • Accumulators and reservoir design • Piping layout and fitting types	L & Ds	PPT, PDF	20 min
(c)	• Complete typical layout of a basic hydraulic system • Flow path from pump to actuator return • Pressure relief and filter placement • System start-up and shutdown procedures • Common failure points in basic layouts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why hydraulic systems require specific fluid viscosity ratings.
(b) Describe the role of the pressure relief valve in a basic layout.
(c) Identify three key components in the typical layout of a hydraulic system.

12. Home Assignment and Follow Up	Prepare a brief note on the specifications of hydraulic oil and the need for clean pipelines, linking this to the protective measures discussed in LP 08.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Specification of hydraulic oil, need for clean pipelines and use of protective ointment while handling oils.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min